DEVOPS PRACTICAL | ARCHITECTURE POSTER

Travel Insurance Claim System

Automated DevOps CI/CD Lifecycle

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THE PROBLEM (in simple words)

Old insurance claims use paper forms and take 15 to 20 days. Manual builds and deployments also cause errors and downtime. We built a digital claim portal and an automated pipeline: push code, test, pack in Docker, deploy and roll back if needed.

15	5	100%	<15 s
tasks done	CI/CD stages	tests pass	auto rollback

APPLICATION ARCHITECTURE - 3 Tiers

1. PRESENTATION
HTML5, CSS3, JavaScript. Customer portal and reviewer dashboard.

→

2. APPLICATION
Java 21 + Spring Boot 3.3.4. REST APIs and validation on port 8081.

→

3. PERSISTENCE
H2 in-memory database. No setup needed and fast for tests.

CI/CD PIPELINE - From commit to live deployment

1 SOURCE Git | GitHub
Developers use feature branches. Code reaches main only through reviewed Pull Requests.

```
git checkout -b feature/claim-validation
git push origin feature/claim-validation
# open Pull Request -> review -> merge to main
```

2 BUILD & TEST Jenkins | Maven | Selenium
Jenkinsfile runs on every commit. JUnit 5 and headless Selenium tests must pass.

```
stage('Build')   { sh './mvnw clean compile' }
stage('Test')    { sh './mvnw test' }  # JUnit + Selenium
stage('Package') { sh './mvnw package -DskipTests' }
```

3 PACKAGE Docker
Multi-stage build keeps the image small. The container runs as a non-root user.

```
FROM maven:3.9.6-eclipse-temurin-21 AS build
FROM eclipse-temurin:21-jre-alpine
USER devops   # non-root, uid 1001
```

4 DEPLOY Ansible
Idempotent playbook replaces the old container with the new version.

```
ansible-playbook -i inventory/hosts.yml \
    playbooks/deploy.yml
# backup: app-previous.jar
```

5 HEALTH CHECK Ansible | curl
After every deployment, the app must answer with HTTP 200 on port 8081.

```
ansible-playbook playbooks/health-check.yml
curl -f http://localhost:8081/index.html
```

PASS - New version is live
Users reach the app on port 8081 with zero manual steps.

FAIL - Automatic rollback
rollback.yml restores app-previous.jar in under 15 seconds.

RESULTS - Before vs After DevOps

Metric	Manual	DevOps	Impact
Claim processing cycle	15-20 days	< 5 minutes	99% faster
Build and packaging	10-15 min	28 seconds	95% faster
Regression testing	2-3 hours	11.4 seconds	Fully automated
Deployment downtime	20-30 min	< 15 seconds	90% less
Rollback recovery	Hours	< 15 seconds	Self-healing

AUTOMATED TESTS

- ✓ 8 MockMvc unit tests (JUnit 5)
- ✓ Claim submission journey
- ✓ Reviewer approval journey
- ✓ Rejection with mandatory reason
- ✓ Form validation (HTTP 400)

Headless Chrome, runs in ~12 s

15-TASK AGILE ROADMAP

Tasks 1-4	Tasks 5-8	Tasks 9-10	Tasks 11-12	Tasks 13-15
Plan, design, Git setup	Branching, Jenkins, Jenkinsfile	Selenium tests + CI	Docker + deployment	Ansible, rollback, release

CONCLUSION

All 15 DevOps tasks done. Code moves from commit to live deployment automatically, with a self-healing rollback.

FUTURE SCOPE

Kubernetes and Helm, Prometheus and Grafana monitoring, SonarQube and OWASP ZAP security scans.